Responsibility Convergence in Organizational Investigations: Structural Dynamics of Blame in Root Cause Analysis

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Abstract

Organizational investigations such as root cause analysis (RCA) are intended to identify systemic causes of failure and support organizational learning. Yet across regulated and safety-critical environments, investigative conclusions recurrently stabilize responsibility at identifiable individuals despite formal commitments to system-oriented explanation. This persistent actor-centered pattern presents a theoretical puzzle: why do investigative systems designed to diffuse responsibility across sociotechnical structures so often concentrate accountability at particular organizational members?

This paper develops a structural explanation for this recurrent stabilization dynamic, termed responsibility convergence. Responsibility convergence refers to the patterned concentration of accountability at identifiable actors when investigative processes translate causal ambiguity into administratively actionable conclusions. The framework specifies how convergence arises from the interaction of three mechanisms: procedural ambiguity, attributional closure pressure, and accountability legibility bias. Under conditions of underdetermined causal reconstruction, organizations face institutional pressure to stabilize responsibility assignments. Identifiable individuals constitute the most interpretable and administratively actionable attribution anchors within accountability infrastructures. As a result, attribution systematically concentrates on actors rather than distributed structural conditions.

By situating investigative convergence within broader organizational responsibility alignment dynamics, the paper clarifies why system-oriented investigative frameworks recurrently produce actor-centered accountability outcomes. The analysis develops a mechanism-based account of convergence, derives theoretical propositions, and outlines implications for organizational learning and accountability system design.

Introduction

Organizational investigations are central mechanisms through which institutions interpret failures

and stabilize responsibility attribution following adverse events. In safety-critical and highly regulated domains, adverse events rarely have a single cause but emerge from interactions among technologies, procedures, and human action. Investigative frameworks such as Root Cause Analysis (RCA) are designed to diffuse responsibility across sociotechnical systems and identify contributing factors across organizational levels rather than individual culpability (Reason, 1990). These approaches reflect a long-standing recognition in safety science and organizational theory that failures in complex systems are distributed phenomena requiring systemic explanation (Perrow, 1984; Vaughan, 1996). Yet organizational failures rarely yield complete causal reconstruction, leaving causal relations structurally indeterminate (Perrow, 1984; Vaughan, 1996).

Beyond their analytic function, investigations also perform a constitutive organizational role. They are the processes through which ambiguous events are rendered intelligible, accountability is assigned, and lessons for future prevention are articulated. Across domains such as safety incidents, compliance failures, and operational disruptions, investigations operate as formalized episodes of organizational sensemaking in which causally underdetermined situations must be translated into administratively actionable accounts that stabilize responsibility (Weick, 1995). Investigative processes are therefore expected simultaneously to preserve systemic explanations and to produce determinate conclusions that support remediation, sanction, and organizational learning. The tension between distributed causality and determinate accountability makes investigative settings analytically revealing sites for examining how responsibility becomes stabilized in practice.

However, across regulated and safety-critical environments, investigative structures frequently stabilize responsibility at identifiable actors such as operators, supervisors, or local decision makers, even when formal frameworks emphasize systemic causation (Reason, 1997; Vaughan, 1996; Dekker, 2007; Hopkins, 2012).

Accident inquiries, audit reviews, and compliance investigations often channel corrective actions toward retraining, procedural adherence, or disciplinary measures directed at specific individuals or roles (Dekker, 2007; Power, 1997).

This convergence pattern persists despite explicit commitments to system-level explanation and organizational learning embedded in contemporary safety and governance frameworks. The recurrent stabilization of accountability at identifiable actors therefore poses a theoretical puzzle: why do investigative systems designed to diffuse responsibility across sociotechnical structures so often stabilize attribution at particular organizational actors?

This paper addresses this question by introducing the concept of responsibility convergence. Responsibility convergence refers to the structural stabilization of accountability at identifiable

actors when investigative processes, structured by accountability infrastructures, translate causal ambiguity into administratively actionable accounts. Rather than reflecting intentional blame or cognitive bias alone, convergence arises from structural properties of investigative reasoning and accountability infrastructures. The analysis develops a mechanism model in which procedural ambiguity, attributional closure pressure, and accountability legibility bias sequentially channel attribution toward actors who provide the most interpretable, documentable, and sanctionable anchors for organizational action.

Existing research has extensively documented the tendency of organizational investigations to attribute responsibility to identifiable individuals rather than distributed structural conditions (Reason, 1997; Vaughan, 1996; Dekker, 2007). However, these accounts primarily describe the empirical pattern of actor-centered attribution without specifying the structural mechanisms through which investigative processes repeatedly generate such outcomes.

By shifting the focus from descriptive accounts of blame patterns to the specification of structural mechanisms, the present framework explains how investigative systems translate causal ambiguity into administratively actionable conclusions. Specifically, the interaction of procedural ambiguity, attributional closure pressure, and accountability legibility bias channels attribution toward identifiable actors even when systemic explanations remain plausible. Situating investigative convergence within broader responsibility dynamics, the paper contributes to organizational theory in three ways. First, it reframes blame in investigations as a predictable structural stabilization of responsibility rather than a deviation from systemic ideals. Second, it integrates attribution theory with accountability and governance perspectives by specifying how responsibility becomes operationally actionable within organizational procedures. Third, it positions responsibility convergence as the investigative stage within a broader research program on responsibility structuring across organizational processes, linking procedural pre-configuration, anticipatory alignment, and post-hoc stabilization of accountability. Responsibility convergence operates at the post-event stage of organizational accountability processes. It complements prior work on anticipatory alignment and procedural judgment in organizational systems by specifying how responsibility becomes stabilized after events within investigative infrastructures. The framework therefore focuses on the post-event phase of organizational accountability processes, where causally ambiguous events must be translated into administratively actionable responsibility assignments.

Responsibility Attribution in Organizational Investigations

This section reviews three bodies of prior work that together establish the conceptual terrain

responsibility convergence addresses: (a) attribution theory and organizational blame research, which explain individual-centered attribution as a cognitive and evaluative process; (b) RCA and organizational investigation research, which specify the procedural context in which attribution occurs; and (c) accountability and governance scholarship, which establishes the institutional conditions under which attribution must satisfy administrative requirements. Responsibility convergence theory extends beyond each of these individually by specifying how their interaction within investigative contexts generates patterned attribution outcomes that none can fully explain alone.

Responsibility attribution in organizations has been examined across attribution theory, accountability research, and organizational blame studies (Heider, 1958; Weiner, 1985; Roberts, 1991). Attribution theory distinguishes between dispositional and situational explanations of events, while organizational research has shown that negative outcomes are often attributed to individuals under conditions of uncertainty or causal complexity (Staw, 1975; Ross, 1977). Accountability scholarship further demonstrates that expectations of evaluation shape decision and justification processes (Tetlock, 1985; Lerner & Tetlock, 1999). Roberts (1991) further identifies how accountability relationships are constituted through the mutual recognition of obligations and answerability, establishing that accountability is not merely a post-hoc evaluative judgment but a relational structure that shapes how actors anticipate and respond to scrutiny. However, these perspectives primarily explain attribution as a cognitive or evaluative process rather than as an outcome structured by investigative procedures and accountability infrastructures.

While existing research on organizational blame has frequently interpreted actor-centered responsibility attribution as a form of scapegoating (Boeker, 1992; Pfeffer & Salancik, 1978), responsibility convergence describes a structurally distinct mechanism. Scapegoating accounts typically emphasize politically motivated blame allocation, symbolic displacement of responsibility, or strategic protection of organizational legitimacy. In contrast, responsibility convergence does not require intentional blame shifting or strategic blame management. Instead, it emerges from the interaction between procedural ambiguity, institutional closure pressures, and accountability infrastructures that privilege administratively legible attribution targets. Under such conditions, responsibility may converge on identifiable actors even in the absence of deliberate scapegoating, as investigators navigate institutional requirements for determinate and documentable causal accounts.

In organizational investigations, procedures such as RCA integrate attributional reasoning with procedural governance (Rooney & van den Heuvel, 2004; Andersen & Fagerhaug, 2006). Investigators must construct causal narratives linking events, actions, and outcomes. Yet

organizational failures rarely yield complete causal reconstruction, leaving causal relations structurally indeterminate (Perrow, 1984; Vaughan, 1996). Event sequences are distributed, nonlinear, and often partially documented. Multiple plausible explanations coexist, and evidentiary completeness is rarely achievable in practice.

This structural indeterminacy produces a closure requirement for accountability stabilization. Investigations cannot remain indefinitely ambiguous; they must stabilize determinate accounts that enable corrective action, regulatory documentation, and organizational learning processes. Reason's (1997) framework for managing organizational accident risk specifies that effective investigation requires tracing causal pathways to their origins within organizational systems, yet simultaneously acknowledges that this aspiration is constrained in practice by the administrative requirements of reporting, documentation, and corrective action assignment that investigations must satisfy.

This tension between systemic aspiration and administrative constraint creates the conditions under which responsibility convergence becomes likely. Responsibility attribution in RCA contexts therefore occurs under conditions of constrained certainty and institutional pressure for determinate outcomes.Within this context, responsibility convergence can be understood as a stabilization dynamic: attribution settles on targets that satisfy interpretive clarity and administrative actionability under ambiguity. Roberts (1991) demonstrates that accountability relationships are structured around the possibilities of answerability and sanctionability that organizational relationships create, while Roberts (2009) elaborates how formal accountability systems generate systematic pressures toward visible, demonstrable attribution that can satisfy governance expectations. Both accounts converge on the observation that accountability infrastructures are organized around the production of interpretable responsibility assignments rather than accurate causal reconstruction. Because organizational accountability expectations therefore orient toward identifiable actors (Roberts, 1991; Power, 1997), investigative closure tends to settle on individuals even when causation is distributed. RCA procedures formalize this convergence dynamic by requiring determinate causal attribution and corrective assignment even when causal reconstruction remains distributed or evidentially incomplete.

Mechanisms of Responsibility Convergence

The analysis identifies three interacting mechanisms that generate responsibility convergence within organizational investigations: procedural ambiguity, attributional closure pressure, and accountability legibility bias.

The argument developed here emphasizes responsibility convergence as a structural dynamic

rather than solely a cognitive attribution bias. Attribution research has demonstrated that individuals often favor actor-centered explanations under uncertainty (Ross, 1977; Heider, 1958). However, convergence dynamics described in this paper operate even in investigative settings where participants are explicitly trained to avoid such biases and to consider systemic causation. The proposed mechanism therefore highlights how features of accountability infrastructures, such as documentation regimes, sanctioning requirements, and expectations of determinate attribution, can channel investigative reasoning toward actor-centered conclusions independently of investigators' individual cognitive tendencies. Responsibility convergence thus emerges from the interaction between evidentiary ambiguity and institutional demands for administratively actionable explanations.

Procedural Ambiguity

Procedural ambiguity refers to indeterminacy in reconstructing causal sequences and accountability relations during post-event investigation (Gephart, 1993; Vaughan, 1996; Weick, 1995). Investigative processes nevertheless require determinate conclusions in order to support organizational action, regulatory reporting, and institutional continuity (March et al., 2000; Dekker, 2007).

Investigations rely on retrospective interpretation, incomplete documentation, and contested narratives. When causal chains cannot be definitively established, multiple attribution pathways remain viable. Responsibility assignments therefore remain indeterminate, generating pressure toward attributional convergence as investigations move toward determinate conclusions.The conversion of evidentiary ambiguity into institutional pressure for closure requires explanation. Rather than tolerating indeterminacy or permitting extended investigation, accountability infrastructures operate under exogenous constraints—such as regulatory reporting deadlines, organizational continuity requirements, and governance expectations for demonstrable resolution—that impose a temporal boundary on interpretive openness (Power, 1997).

These constraints are structural properties of accountability systems rather than the result of individual choices: they apply regardless of investigators' cognitive tolerance for ambiguity or their commitment to systemic explanation. Ambiguity therefore does not remain neutral but is transformed by the institutional frame of investigation into pressure for determinate attribution.

Attributional Closure Pressure

Attributional closure pressure arises from this requirement. As ambiguity persists, pressure increases to select an explanation that is both defensible and operationally actionable. Investigative closure therefore transforms open-ended causal interpretation into determinate attribution outcomes capable

of supporting organizational documentation, remediation, and governance processes.

As investigative processes move toward resolution, closure pressure operates by structuring the selection environment of attribution. Among multiple plausible causal accounts, investigators are driven to privilege explanations that can sustain organizational scrutiny and support actionable decisions. This process does not necessarily resolve causal ambiguity; rather, it constrains which interpretations remain institutionally viable. Closure pressure therefore narrows the interpretive field, channeling attribution toward determinate outcomes that can be justified, documented, and operationalized within organizational procedures.

Crucially, closure pressure does not select among attribution targets randomly. Instead, the selection environment is structured by the properties of available attribution anchors within the accountability infrastructure. Collective units, distributed processes, or systemic design features are frequently not represented as discrete, nameable entities within the evidentiary records and governance procedures upon which investigators rely. Role-based attributions may exist within formal accountability structures, but roles must ultimately be enacted by individuals whose specific actions can be documented and subjected to formal sanction. Systemic attributions, while analytically compelling, often require causal specification at a level of abstraction that resists direct operationalization within corrective action frameworks, disciplinary procedures, and regulatory filings. Individual actors, by contrast, possess properties that make them administratively tractable: they are nameable, their actions can be traced within procedural records, and their behavior can be subjected to remediation through established governance instruments. Closure pressure therefore does not produce arbitrary attribution but instead amplifies the selection advantage of individual actors as attribution targets within existing accountability architectures.

Accountability Legibility Bias

Accountability legibility bias refers to the structural tendency of accountability infrastructures to privilege attribution targets that can be clearly represented within documentation, reporting systems, and sanctioning procedures. Within many organizational investigative contexts, individuals more readily satisfy these criteria than distributed processes, informal coordination failures, or systemic design limitations. Individuals can be named, traced within procedural records, and linked to specific actions within documentary chains of evidence. As a result, individual actors frequently become the most administratively tractable units through which responsibility can be articulated and operationalized. Importantly, this tendency does not necessarily arise from investigator intent or cognitive bias alone, but from institutional requirements that accountability determinations be

representable within formal governance procedures.

Accountability systems tend to orient toward identifiable agents as attribution targets (Latour, 1987; Roberts, 2009). This orientation produces an accountability legibility bias: responsibility is more readily assigned to entities that are traceable, documentable, and sanctionable within organizational records and governance processes. While prior accountability research has emphasized the role of identifiable agents, the present analysis specifies how legibility operates as a structural selection mechanism during investigative closure. Individual actions typically satisfy these criteria because they appear discrete and normatively evaluable, whereas systemic causes remain diffuse, distributed, and structurally embedded. Because accountability infrastructures privilege attribution targets that can be clearly recorded and acted upon, individuals become the most legible recipients of responsibility under conditions of attributional closure.

Closure pressure alone, however, does not determine where responsibility ultimately stabilizes. Investigative processes must produce conclusions that are not only determinate but also organizationally actionable. Closure therefore operates through attribution anchors that satisfy documentation, remediation, and accountability requirements, directing attribution toward actors that can be clearly specified within organizational records and corrective frameworks. Responsibility convergence thus reflects not merely a bias toward individual blame but the structural requirement that accountability systems render causality governance-compatible under conditions of evidentiary constraint.

Convergence Dynamics

These mechanisms operate in interaction rather than isolation. Procedural ambiguity structurally intensifies closure pressure by expanding the range of plausible causal interpretations that must ultimately be resolved. Closure pressure, in turn, activates or amplifies legibility bias by structuring the selection environment in which attribution targets are evaluated. Under these conditions, responsibility is more likely to stabilize around attribution anchors that are traceable, documentable, and sanctionable within organizational accountability systems. Responsibility convergence therefore emerges as a structural outcome of investigative reasoning conducted under accountability constraints, rather than as the product of intentional blame allocation or purely cognitive attribution bias.

These mechanisms operate sequentially within investigative processes. Procedural ambiguity expands the range of plausible causal interpretations that must be reconstructed following an adverse event. This expansion generates pressure for closure because investigative systems must ultimately

stabilize determinate accounts that support regulatory reporting, corrective action, and organizational learning. Under conditions of closure pressure, attribution tends to converge on actors whose actions are most legible within organizational accountability infrastructures. Legibility bias therefore operates as the selection mechanism through which closure pressure resolves the interpretive indeterminacy generated by procedural ambiguity.

This dynamic also reflects broader organizational tendencies to align responsibility with identifiable accountability targets within governance and accountability infrastructures. Organizational systems must translate causal explanations into administratively actionable outcomes that can support documentation, remediation, and sanctioning processes. As a result, responsibility tends to converge on actors whose actions can be clearly specified within organizational records and accountability procedures. Investigative convergence thus constitutes a localized manifestation of broader responsibility alignment pressures within regulated accountability systems.

Importantly, the three mechanisms reinforce one another during investigative reasoning. Procedural ambiguity expands the range of plausible causal interpretations, thereby intensifying the institutional pressure for attributional closure. Closure pressure then amplifies the practical consequences of legibility bias by increasing the urgency with which investigators must identify administratively viable attribution targets. Conversely, the existence of legible accountability anchors—typically identifiable individuals—provides investigators with a readily available pathway for resolving closure pressure. Responsibility convergence therefore emerges not from any single mechanism alone but from their interaction within investigative systems tasked with transforming causal ambiguity into actionable accountability assignments.

Illustrative Case: Volkswagen Emissions Investigation

The 2015 Volkswagen emissions scandal provides an illustrative case of how responsibility convergence dynamics operate in post-event organizational investigations. Following the discovery that Volkswagen had installed defeat-device software in approximately eleven million diesel vehicles to manipulate emissions test results, regulatory authorities and internal investigators faced the task of constructing a determinate accountability narrative from a causally distributed organizational failure. The development and deployment of the defeat device involved coordinated decisions across engineering, management, and compliance functions over several years, implicating multiple organizational levels and cross-functional coordination structures (Ewing, 2017).

Despite this distributed causal structure, investigative and legal processes concentrated accountability on a limited set of identifiable engineers and executives. Proceedings by the United

States Department of Justice resulted in criminal charges against specific Volkswagen engineers and executives involved in the defeat-device scheme (U.S. Department of Justice, 2017). Volkswagen's internal investigation, conducted by the law firm Jones Day, similarly attributed primary responsibility to a small group of engineers in the diesel development division (Ewing, 2017). The broader organizational structures, resource allocation pressures, and compliance culture that enabled sustained regulatory circumvention received comparatively limited systematic scrutiny in official accountability outputs relative to the attribution of individual responsibility.

This pattern reflects the sequential operation of the three convergence mechanisms proposed in this framework. Procedural ambiguity was substantial: causal reconstruction required tracing underdocumented decisions across distributed organizational units over an extended period. Attributional closure pressure intensified as regulatory deadlines, shareholder demands, and reputational damage created urgent institutional requirements for determinate conclusions. Accountability legibility bias therefore directed attribution toward identifiable engineers and executives whose actions could be rendered traceable, documentable, and legally sanctionable within existing governance frameworks. In this way, the investigative process satisfied the administrative requirement for actionable accountability even as systemic organizational conditions remained analytically underspecified.

The Volkswagen case also illustrates the learning consequences implied by the convergence model. Post-investigation corrective actions emphasized compliance restructuring, personnel changes, and enhanced monitoring systems rather than fundamental redesign of the organizational coordination and incentive structures that had sustained the concealment. Actor-centered attribution thus achieved regulatory closure while limiting the depth of systemic organizational learning, consistent with the framework's prediction that responsibility convergence can constrain the scope of post-investigation learning.

This case is not intended as a comprehensive causal analysis of the Volkswagen scandal but rather serves as an illustrative process tracing of the proposed convergence mechanisms within a paradigmatic large-scale organizational investigation. The analysis highlights how each proposed mechanism operated sequentially within the investigative sequence, distinguishing empirical observations about documented accountability outcomes from theoretical interpretation of the structural conditions shaping them. The case therefore provides plausibility evidence for the convergence model rather than definitive empirical proof and is intended to support theoretical elaboration rather than causal inference about the Volkswagen organization itself.

Propositions

The convergence model yields three theoretical propositions.

Proposition 1. Responsibility convergence arises from the sequential interaction of procedural ambiguity, attributional closure pressure, and accountability legibility bias rather than from any single mechanism operating independently.

Procedural ambiguity expands the field of viable attribution pathways, generating indeterminate responsibility assignments. This indeterminacy activates closure pressure, as investigative processes face institutional demands to stabilize attribution into defensible and actionable conclusions. Closure pressure, however, does not determine where attribution settles; it determines that attribution must settle. Legibility bias then resolves this selection problem by directing attribution toward actors who are most traceable, documentable, and sanctionable within organizational accountability infrastructures. Responsibility convergence therefore requires the joint activation of all three conditions. Ambiguity without closure pressure leaves indeterminacy unresolved; closure pressure without legibility bias does not systematically channel attribution toward individuals; and legibility bias without closure pressure lacks the institutional force required to override distributed causal accounts. The sequential coupling of these mechanisms produces recurrent actor-centered attribution outcomes in organizational investigations.

Empirically, this proposition could be evaluated through comparative analysis of investigative reports that vary in evidentiary completeness and institutional closure pressure. For example, content analysis of RCA reports could code for indicators corresponding to each mechanism: the degree of causal indeterminacy reflected in the evidentiary record, the presence of time pressure or external regulatory oversight as proxies for closure pressure, and the concentration of corrective actions on identifiable actors as an indicator of legibility bias activation. Convergence intensity could then be assessed through the distribution of attribution targets in investigative conclusions (individual versus systemic), the proportion of corrective actions directed at individual behavior versus structural redesign, and the alignment between evidentiary complexity and attribution breadth across investigative cases.

This configurational claim is falsifiable in principle. Convergence should not be observed when any one of the three conditions is absent: in settings where investigations face no institutional pressure for determinate conclusions (ambiguity without closure pressure), where closure pressure is present but accountability infrastructures provide structurally legible systemic anchors (closure pressure without individual legibility advantage), or where legibility bias exists but institutional commitments to systemic explanation are formally enforced through investigation design.

Proposition 2. Greater responsibility convergence toward identifiable actors is negatively associated with the depth of systemic learning in post-investigation corrective processes.

When responsibility converges on identifiable actors, corrective processes tend to target behavior, compliance, or local decision quality rather than underlying structural conditions. Organizational responses frequently emphasize retraining, monitoring, or disciplinary measures directed at specific individuals or roles rather than redesigning procedures, resource configurations, or coordination structures that contributed to failure. Actor-centered attribution directs investigative attention toward proximal actions, limiting examination of the distributed system interactions underlying failure. As a result, investigations may achieve attributional closure without generating deeper organizational learning about structural vulnerabilities, shifting post-investigation adaptation toward local correction rather than systemic reconfiguration even when formal learning mechanisms are present (March et al., 2000).

Empirically, this proposition could be evaluated by coding the distribution of corrective action targets in post-investigation reports. Corrective actions directed at individual behavior, compliance, or retraining would indicate shallow learning, whereas actions targeting structural redesign, process reconfiguration, coordination structures, or resource allocation would indicate deeper systemic learning. Convergence intensity could be operationalized as the proportion of attribution statements directed at identifiable individuals or roles, while systemic learning depth could be indexed by the proportion of corrective actions directed at structural conditions. A negative association between these measures would provide empirical support for Proposition 2 and could be examined through systematic content analysis of investigative reports across regulated industries.

Proposition 3. Responsibility convergence weakens when accountability infrastructures distribute sanctionability across collective units rather than concentrating it on identifiable actors.

Convergence dynamics depend not only on causal complexity but also on the institutional structure of accountability. In settings characterized by shared liability regimes, systemic transparency, and reduced individual sanction intensity, closure pressure does not necessarily translate into actor-centered attribution because accountability infrastructures themselves provide legible systemic anchors. When sanctionability is distributed across collective units, individuals no longer constitute the most administratively efficient stabilization point, and attribution may remain more diffuse and structurally anchored. Responsibility convergence therefore varies across institutional designs, and mitigating it requires redesigning accountability infrastructures rather than revising investigative

procedures alone.

Empirically, distributed sanctionability could be operationalized by coding accountability documentation for the unit of sanction assignment, such as individual, role, team, or organizational level, alongside indicators of shared liability regimes such as collective settlement structures, joint corrective action plans, or explicit attribution of systemic rather than individual causes in official findings. Variation in convergence outcomes across accountability systems with differing sanctionability structures would provide comparative evidence for this proposition.

Distributed accountability architectures are most likely to attenuate convergence when they combine three properties: sanctioning procedures operating at the level of collective units rather than individuals, documentation systems that render systemic conditions as legible within governance procedures as individual actions, and investigative protocols that require structural attribution before individual attribution is considered. When these conditions are only partially present, for example where collective sanctions exist but evidentiary records remain individually organized, convergence attenuation may be incomplete, producing hybrid attribution patterns combining individual and systemic elements.

Scope Conditions

Although responsibility convergence is proposed as a general structural tendency in organizational investigations, the intensity of convergence is expected to vary across institutional and situational conditions. Four scope conditions are particularly relevant to the framework's domain of application.

First, event severity and public visibility amplify convergence by intensifying attributional closure pressure. Adverse events that attract regulatory scrutiny, media attention, or public accountability demands generate institutional requirements for rapid and determinate explanations. Under such conditions, investigators face heightened pressure to produce conclusions that are administratively defensible and publicly communicable, concentrating attribution on the most legible available targets. High-visibility events therefore represent contexts in which convergence dynamics are expected to be particularly strong.

Second, power asymmetry among potential attribution targets modulates the direction of convergence. Where organizationally powerful actors are insulated from attribution through structural protections, such as legal representation, hierarchical authority, or reputational buffering, convergence may redirect toward lower-power front-line actors who satisfy legibility criteria but lack the institutional resources to resist attribution. Conversely, in settings where organizational accountability norms strongly protect front-line workers, such as just culture frameworks in safety-

critical industries, convergence pressure may remain less resolved or redirect toward managerial roles. Power asymmetry therefore shapes which identifiable actors become the terminal targets of convergence dynamics.

Third, evidentiary density may paradoxically intensify convergence in highly documented environments. When electronic records, algorithmic audit trails, or comprehensive logging systems generate dense individual-level documentation, the evidentiary field available to investigators is already organized around discrete actor actions. This pre-structuring of the evidentiary record reduces interpretive space for systemic attribution and amplifies legibility bias by making individual action sequences the most traceable and recoverable elements of the causal record. Highly documented organizational environments may therefore exhibit stronger convergence precisely because rich individual-level evidence foregrounds actor-level explanations during investigative reconstruction.

Fourth, investigation structure, specifically whether investigations are conducted internally or by external bodies, modulates closure pressure and legibility bias in different ways. Internal investigations may face stronger organizational pressures to avoid convergence toward powerful insiders or to redirect attribution toward lower-level actors for self-protective reasons. External investigations, by contrast, may face stronger regulatory expectations for determinate and publicly defensible attribution. These structural differences generate distinct convergence profiles that the framework predicts would be observable through comparative analysis of internal versus external investigative outcomes across analogous adverse events.

These scope conditions do not restrict responsibility convergence to a single organizational type or investigative setting. Rather, they specify institutional and situational parameters that modulate convergence intensity and identify the comparative contexts in which the framework's predictions can be most precisely evaluated.

Theoretical Contributions

The responsibility convergence framework contributes to organizational theory in three ways.

First, it reframes the concentration of blame as a structural convergence process rather than as the outcome of intentional attributional choice. Actor-centered attribution emerges from the structural conditions of investigation, particularly the interaction of ambiguity, closure pressure, and legibility constraints, rather than primarily reflecting individual attributional bias.

Second, the framework integrates attribution theory with accountability and governance literatures by specifying how procedural and institutional requirements shape attribution outcomes

under conditions of causal ambiguity. While attribution research has largely emphasized cognitive and evaluative processes, the convergence framework highlights how accountability infrastructures and investigative procedures structure the range of administratively viable attribution outcomes.

Third, the framework situates investigative attribution within broader organizational responsibility alignment dynamics. Organizational accountability systems tend to orient responsibility toward identifiable actors, and RCA conclusions reflect how localized investigative closure interacts with these wider alignment pressures. Responsibility convergence therefore complements anticipatory alignment dynamics by specifying how responsibility becomes stabilized after events within accountability systems.

This perspective clarifies why system-oriented investigative frameworks frequently produce actor-centered outcomes. Convergence should not be understood as a deviation from investigative ideals but as a predictable structural effect of ambiguity, closure pressure, and legibility constraints operating within accountability infrastructures.

Implications for Organizational Learning and Accountability Design

Responsibility convergence has implications for both organizational learning and the design of accountability infrastructures. When investigations concentrate responsibility on identifiable individuals, systemic vulnerabilities may remain underdiagnosed, and corrective actions may emphasize retraining or disciplinary measures rather than structural redesign. Over time, such patterns may reproduce organizational risk and weaken perceptions of procedural fairness. This outcome aligns with the framework's prediction that responsibility convergence can reduce the depth of systemic learning by directing corrective attention toward proximate actors rather than distributed structural conditions.

Mitigating convergence requires interventions that address the structural mechanisms underlying the process. Reducing procedural ambiguity through richer data capture and improved temporal reconstruction of events may expand the interpretive space available during investigations. Designing investigative protocols that tolerate provisional conclusions or iterative analysis may reduce attributional closure pressure. Enhancing the representation of systemic causal pathways within investigative documentation may counteract legibility bias by making distributed causes more interpretable and actionable within accountability systems.

Such interventions are not intended to eliminate individual responsibility. Rather, they aim to balance responsibility across organizational levels so that investigative conclusions more closely align with the distributed causal structures that characterize failures in complex sociotechnical

systems.

Discussion: Investigative Closure and Responsibility Alignment

Organizational investigations translate ambiguous causal structures into stabilized responsibility assignments under accountability constraints. In regulated environments, where actions must be documented, justified, and auditable, attributional legibility becomes a central organizing principle of investigative reasoning. Responsibility convergence therefore reflects how investigative closure processes align with broader organizational tendencies to orient accountability toward identifiable actors.

Although developed in the context of RCA, responsibility convergence is not unique to formal investigations. Similar convergence dynamics arise whenever organizations must translate causally underdetermined events into administratively actionable accounts, including performance evaluation, audit and compliance processes, and post-hoc decision justification. Across these contexts, distributed or structurally conditioned outcomes are frequently condensed into determinate conclusions that justify intervention. Organizational constraints, coordination breakdowns, or resource dependencies are often reframed as issues of individual competence, procedural adherence, or judgment. In each case, responsibility convergence reflects a shared structural requirement that accountability be rendered identifiable and operational within organizational systems. The same pressures recur: ambiguous causal reconstruction generates demands for closure, while accountability infrastructures privilege attribution targets that are traceable, recordable, and sanctionable. RCA investigations therefore provide an analytically visible setting in which this broader organizational tendency becomes concentrated and observable.

From this perspective, RCA processes do not merely discover causes; they also structure responsibility through interpretive stabilization. The concentration of attribution on individuals emerges from the interaction of ambiguity, closure pressure, and legibility bias within investigative contexts. These dynamics situate investigative blame within broader organizational patterns that orient responsibility toward identifiable accountability targets.

Responsibility convergence thus illuminates how organizations transform distributed causality into administratively actionable attribution. Investigative conclusions represent stabilized points of responsibility alignment under conditions of uncertainty. This stabilization dynamic also shapes organizational sensemaking and learning, as attributional closure influences which causal pathways remain visible and actionable in subsequent corrective processes (Weick, 1995; March et al., 2000). Whereas cognitive friction identifies how procedural structures pre-configure localized

judgment during task execution, responsibility convergence explains how investigative infrastructures subsequently stabilize responsibility attribution after events, locating responsibility alignment across the temporal arc of organizational action.

Responsibility convergence may therefore be understood as one phase within a broader temporal structuring of accountability, in which responsibility is shaped before action, stabilized during investigation, and subsequently embedded within organizational memory.

The relationship between responsibility convergence and anticipated accountability convergence (AAC) also invites examination of how AI-mediated accountability systems may intensify convergence dynamics across both phases. AAC describes how algorithmic performance monitoring systems pre-configure accountability expectations by rendering certain actor behaviors selectively visible within evaluation infrastructures. When such systems generate the documentary records upon which RCA investigators subsequently rely, the legibility infrastructure of post-event investigation is itself shaped by the pre-event visibility structures established through algorithmic monitoring. Responsibility convergence may therefore not only follow but also be structurally amplified by the prior operation of anticipatory accountability systems that selectively encode individual actions as the most traceable and recoverable elements of organizational activity.

This amplification dynamic suggests that as organizations increasingly rely on algorithmic systems for performance evaluation and compliance monitoring, the legibility bias mechanism identified in responsibility convergence may become structurally intensified. When AI-generated logs, dashboards, and audit trails constitute the primary evidentiary base for causal reconstruction, accountability legibility bias becomes embedded within the technological infrastructure itself rather than arising solely from investigative reasoning. Investigators operating under closure pressure encounter an evidentiary field already organized around individual action sequences, reducing the interpretive resources available for systemic attribution and further concentrating convergence at the actor level. Examining how AAC-mediated algorithmic systems shape the documentary conditions of RCA investigations therefore represents a productive direction for both empirical and theoretical extension of the convergence framework.

Future research could empirically examine responsibility convergence by analyzing patterns of attribution across formal investigative reports in regulated environments. Comparative analysis of RCA documents, audit findings, or compliance investigations could assess whether greater procedural ambiguity and closure pressure are associated with increased concentration of responsibility at identifiable actors. Convergence may be operationalized through the distribution of attribution targets in investigative conclusions, corrective action assignments, and documentation

practices. Such empirical work would allow systematic evaluation of when convergence intensifies, when it weakens, and how institutional accountability architectures shape attribution outcomes across organizational contexts.

Conclusion

This paper introduces responsibility convergence as a structural explanation for the tendency of organizational investigations to attribute failures to individuals despite system-oriented intentions. By identifying procedural ambiguity, attributional closure pressure, and accountability legibility bias as interacting mechanisms, the framework clarifies how investigative processes translate causal indeterminacy into concentrated attribution outcomes.

Situating investigative convergence within broader organizational responsibility alignment dynamics highlights how RCA outcomes reflect both localized investigative closure and wider accountability structures. Recognizing convergence dynamics contributes to organizational accountability theory and informs the design of investigative systems that more accurately represent distributed causation within complex sociotechnical environments.

Responsibility convergence therefore provides a framework for understanding how organizations stabilize responsibility under conditions of uncertainty. The framework clarifies the structural mechanisms through which investigative closure concentrates attribution on identifiable actors. In doing so, it contributes to ongoing research on organizational accountability, investigation design, and learning from failure in complex sociotechnical systems.

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